Lecture 4 - Instance-Based Learning

Instances are often defined as vectors with multiple attributes. The *p*-th attribute of the *i*-th instance is defined as x_{ip}.

If the *p*-th attribute is **numeric**, then the distance between instances x_i and x_j at the *p*-th attribute is

$$d_p(x_i, x_j) = \frac{|x_{ip} - x_{jp}|}{\text{range of } p}$$

If the *p*-th attribute is **discrete**, then the distance between instance x_i and x_j at the *p*-th attribute is 1 if the attributes match, else 0.

Minkowski Distance

The Minkowski distance describes the distance between two instances x_i and x_j:

$$d(x_i, x_j) = \sqrt[m]{\sum_p d_p(x_i, x_j)^m}$$

where m represents the order. properties of Minkowski distance:

- **positive definiteness**: $d(x_i, x_i) = 0$ and $d(x_i, x_j) > 0$ if $i \neq j$

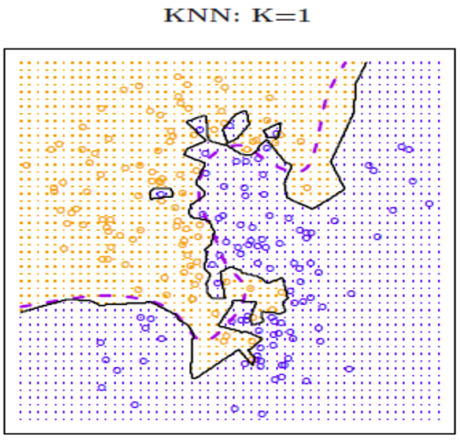
- **symmetry**: $d(x_i, x_j) = d(x_j, x_i)$

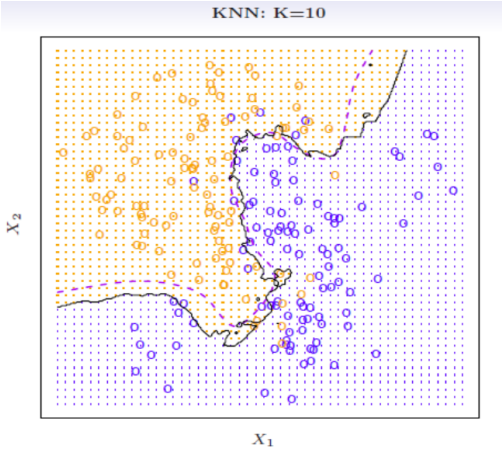
- **triangle inequality**: $d(x_i, x_j) \leq d(x_i, x_k) + d(x_k, x_j)$

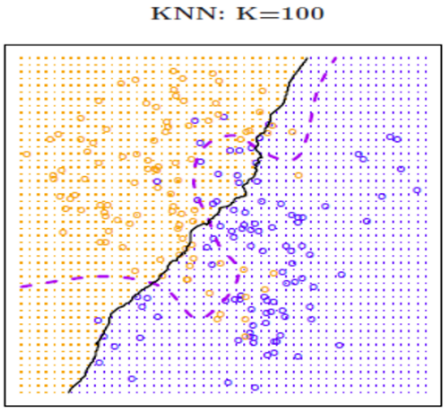
Any distance measure satisfying these properties is a **metric**.

k-NN Classifier

Estimate the class of an instance using the *k* nearest neighbours of the instance (non-linear classifier):

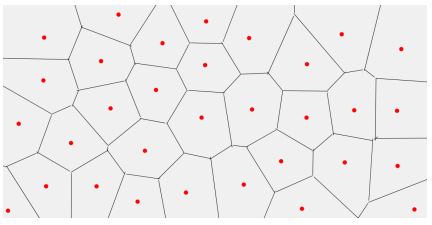






K-NN NOTE: CONTINUOUS VARIABLES SHOULD BE NORMALIZED

Voronoi Diagram:



Advantages of k-NN
- can estimate complex class borders
- learns very quickly
- robust to training noice
- easy to explain and understand

Disadvantages of k-NN
- needs to store all the data
- O(n) classification
- performance decreases with irrelevant variables

k-NN Regression

Predict the class y of an instance:

$$f(x) = \frac{1}{k} \sum_{(x_i, x_j) \in NN} y_i$$

The value of k controls the complexity of the regression curve (as seen above).

Condensing

Aim is to decrease the number of classification instances. We define set S as the set of instances needed to classify new instances.

S is **decision boundary consistent** if the instances in S impose a decision boundary identical to the boundary of the entire training set (the shape can be different, as long as the classification is the same).

S is a **minimum consistent set** if it is the smallest subset of the original data that correctly classifies the entire training set.

k-NN can be combined with decision trees. First classify using the decision tree, but only for a few attributes. Once the number of attributes left has been lowered, classify using *k*-NN.

CNN Classifier

Incremental condensing algorithm. Note that is neither minimal or decision boundary consistent.

1) Initialize set S with a single training instance.

2) Classify all remaining instances using S. If an instance is incorrectly classified put it in S.

3) Repeat until data set is empty of S has reached max capacity.

Edited NN Classifier

Made to improve performance of *k*-NN when there is a lot of noice. It averages about 20-40% condensing, and leaves smoother decision boundaries.

1) Initialize set S as all training data.

2) Each instance is removed from S if it does not agree with its *k* (usually *k* = 3) nearest neighbours in S.